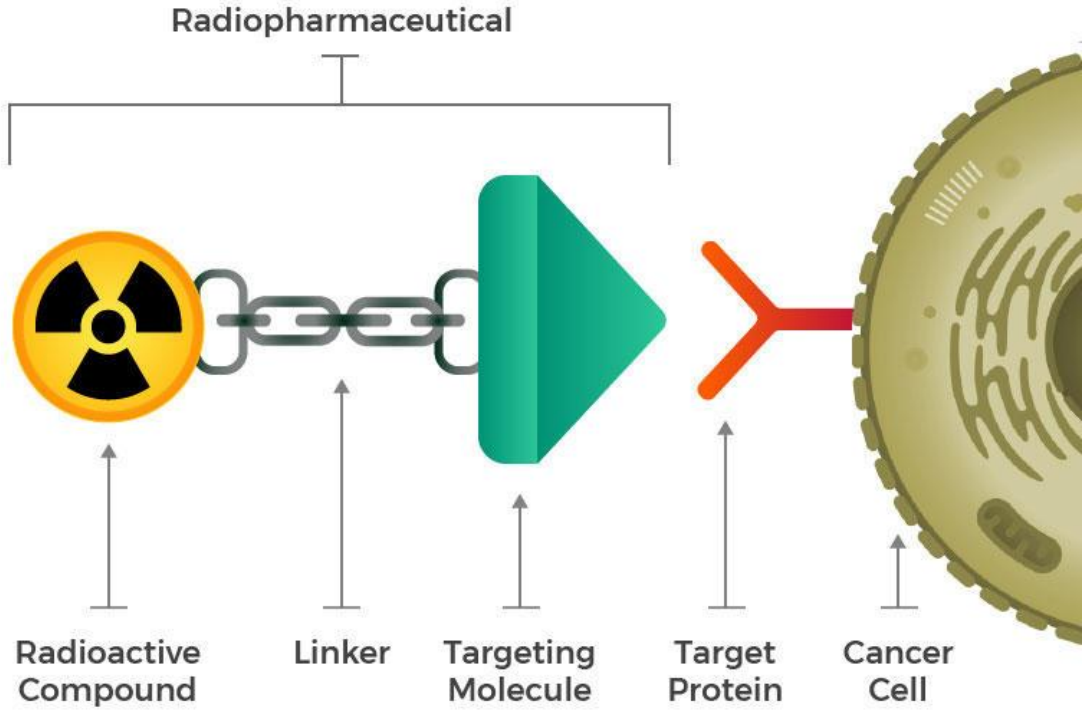
SH1011 Modern fysik, HT 2022

Föreläsning 12:
Repetition
Supraledning
Atomkärnans egenskaper.
Nuklider och isotoper.
Enheten u.
Bindningsenergi.
Stark växelverkan - som binder atomkärnan.
Stabilitet och radioaktivitet.
Alfasönderfall.
Betasönderfall.
Gammasönderfall.
Naturlig radioaktivitet.
Aktivitet och halveringstid.
Hur fungerar MR?



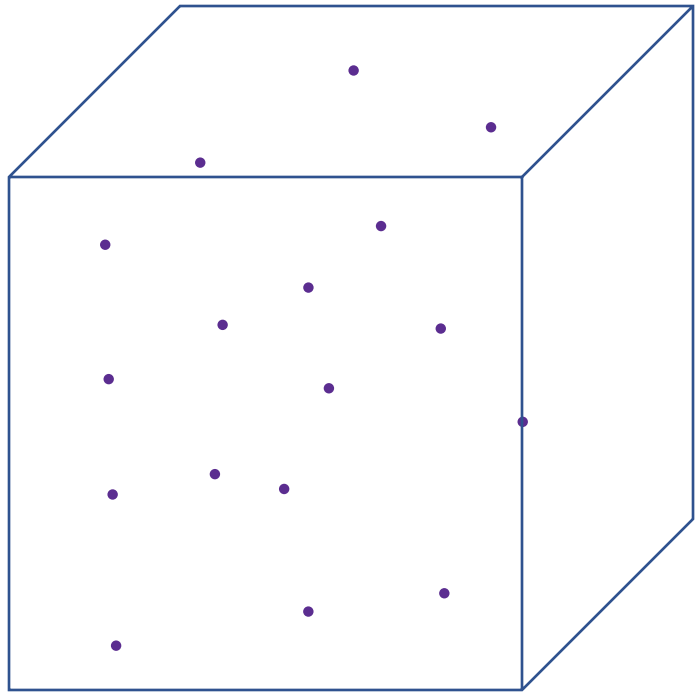
Mats Danielsson

Professor, Institutionen för fysik, KTH

mats.danielsson@mi.physics.kth.se

Young and Freedman: University Physics

Tillståndstäthet för fri elektrongas



Volym V

$$g(E) = \frac{(2m)^{\frac{3}{2}} V}{2\pi^2 \hbar^3} E^{\frac{1}{2}}$$

”Antal elektrontillstånd per energienhet”

Dimension: energi^{-1}

0.1 mm3 metal I

Hur manga energinivåer finns från 0.5 meV till 25 meV?

Integrera tillståndstätheten

Ger 1.79*10^16 tillstånd

För en vit dvärg kan energin vra 10 MeV

Med volym som Jorden blir detta totalt 6 x 10^59

Fermi-Dirac-fördelningen

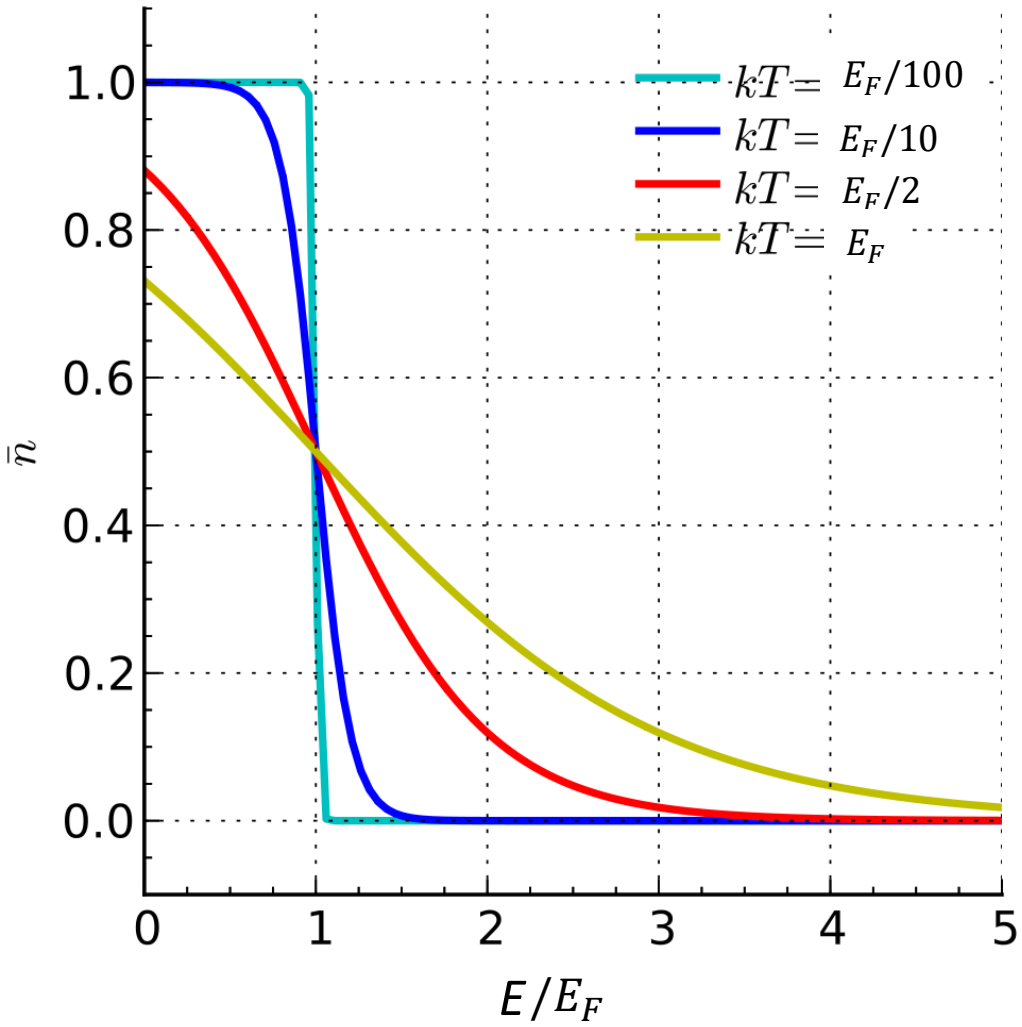
Vad är sannolikheten att hitta en partikel i ett tillstånd med energi E?

Gäller för **fermioner**

$$f(E) = \frac{1}{e^{\frac{E-E_F}{kT}}+1}$$

$k = 1{,}38 \cdot 10^{-23} J/K$ Boltzmanns constant

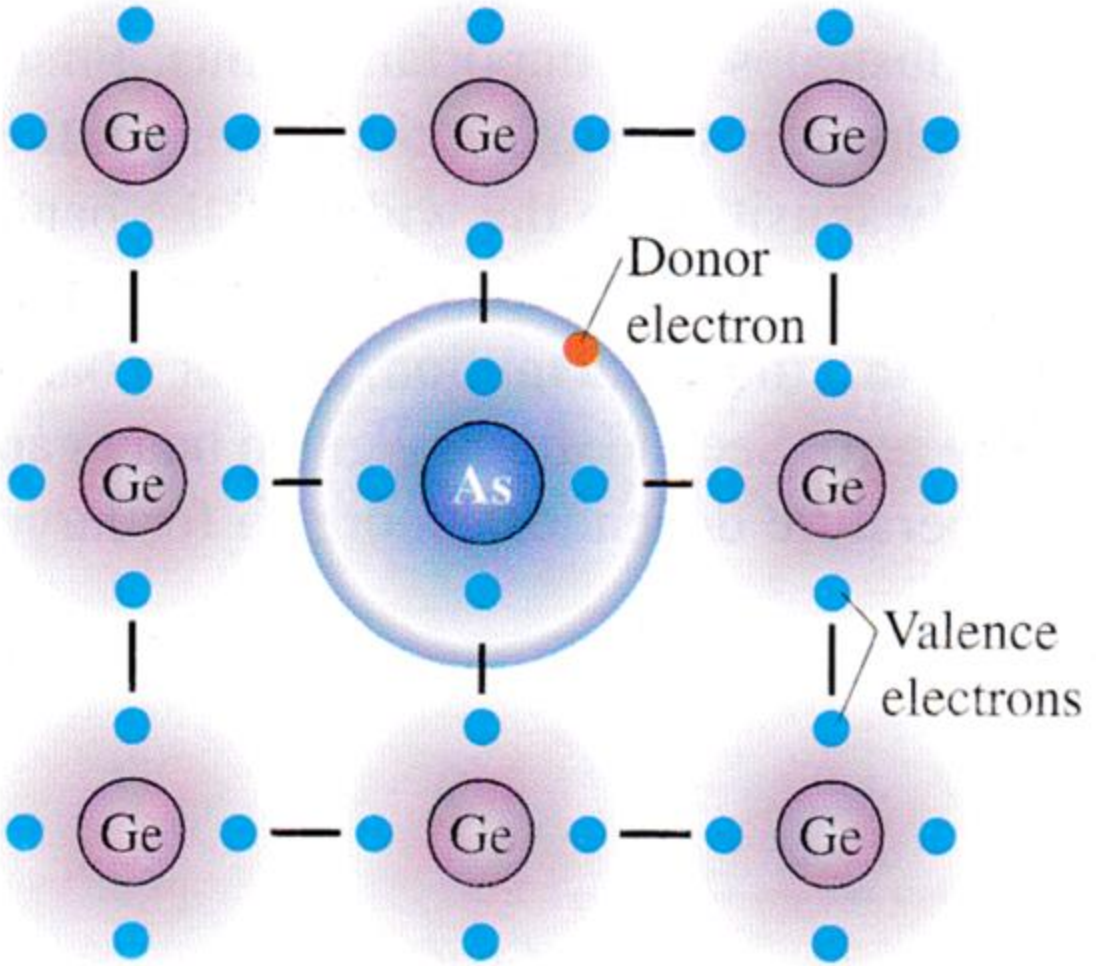
E_F= Ferminivån



Elektrontäthet per energi $= g(E)f(E)$

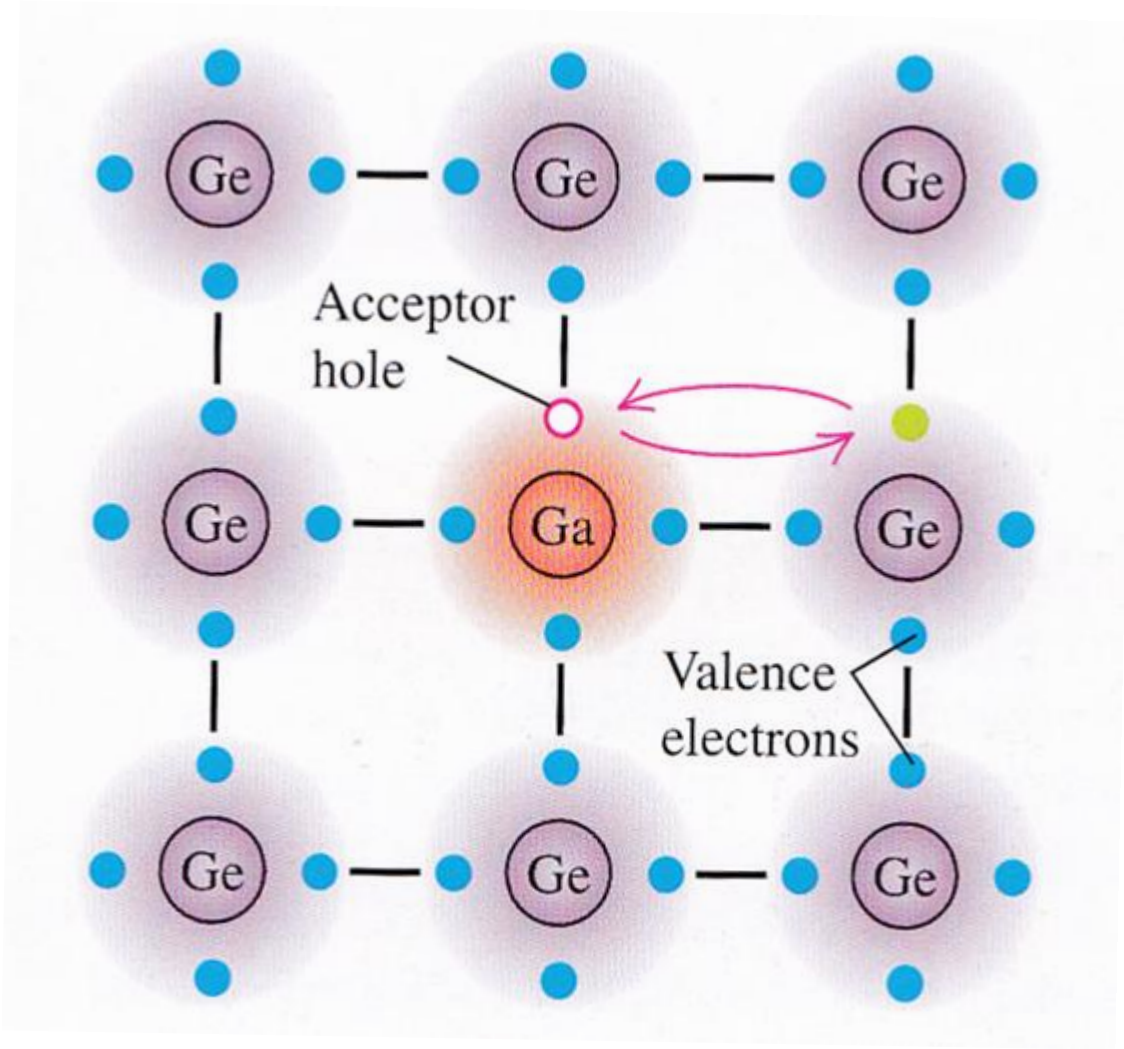
https://en.wikipedia.org/wiki/Fermi%E2%80%93Dirac_statistics#/media/File:FD_e_mu.svg

N-dopad halvledare



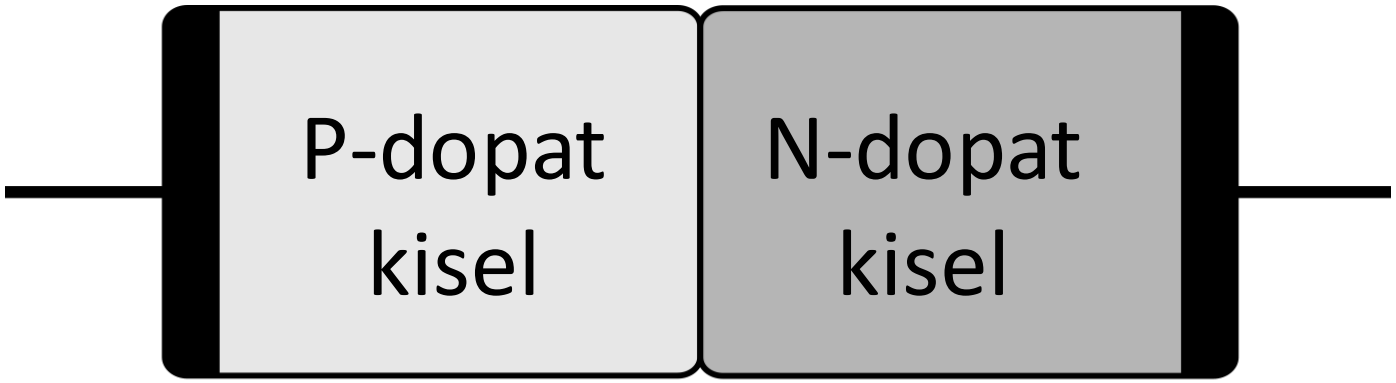
Young and Freedman: University Physics

P-dopad halvledare



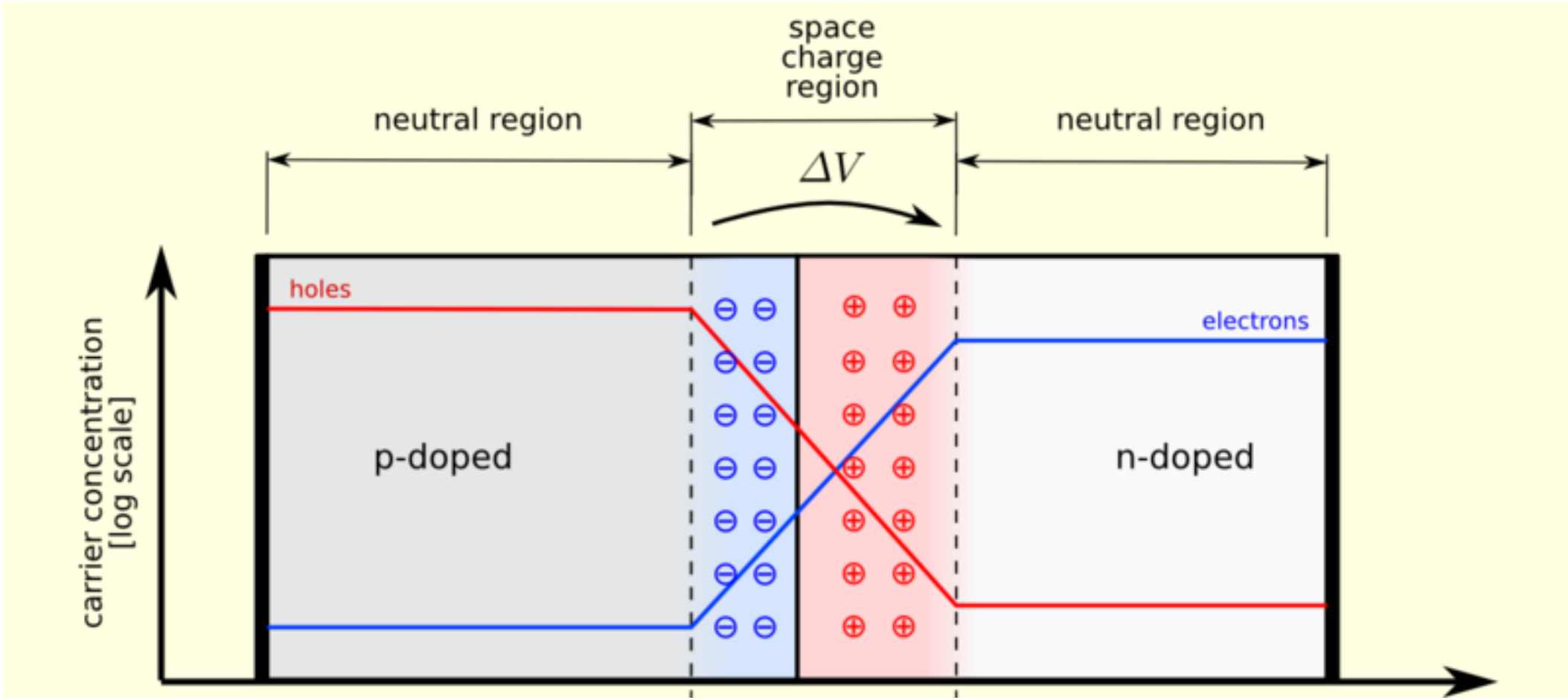
Young and Freedman: University Physics

Diod (PN-övergång)



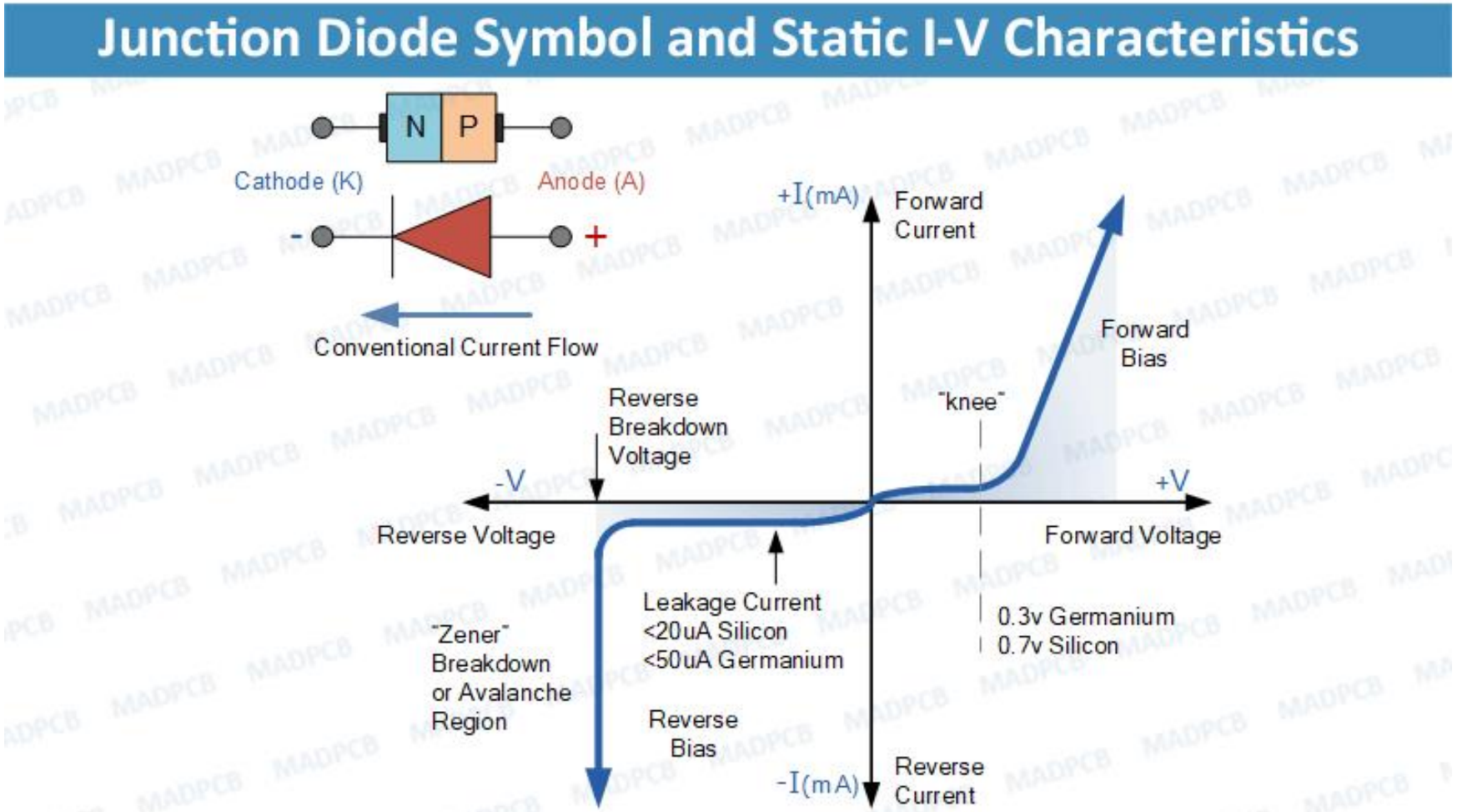
Raffamaiden https://en.wikipedia.org/wiki/P%E2%80%93n_junction#/media/File:PN_diode_with_electrical_symbol.svg

Diod (PN-övergång)



Ström och spänning genom en diod

$$I = I_S(e^{\frac{eV}{kT}} - 1)$$

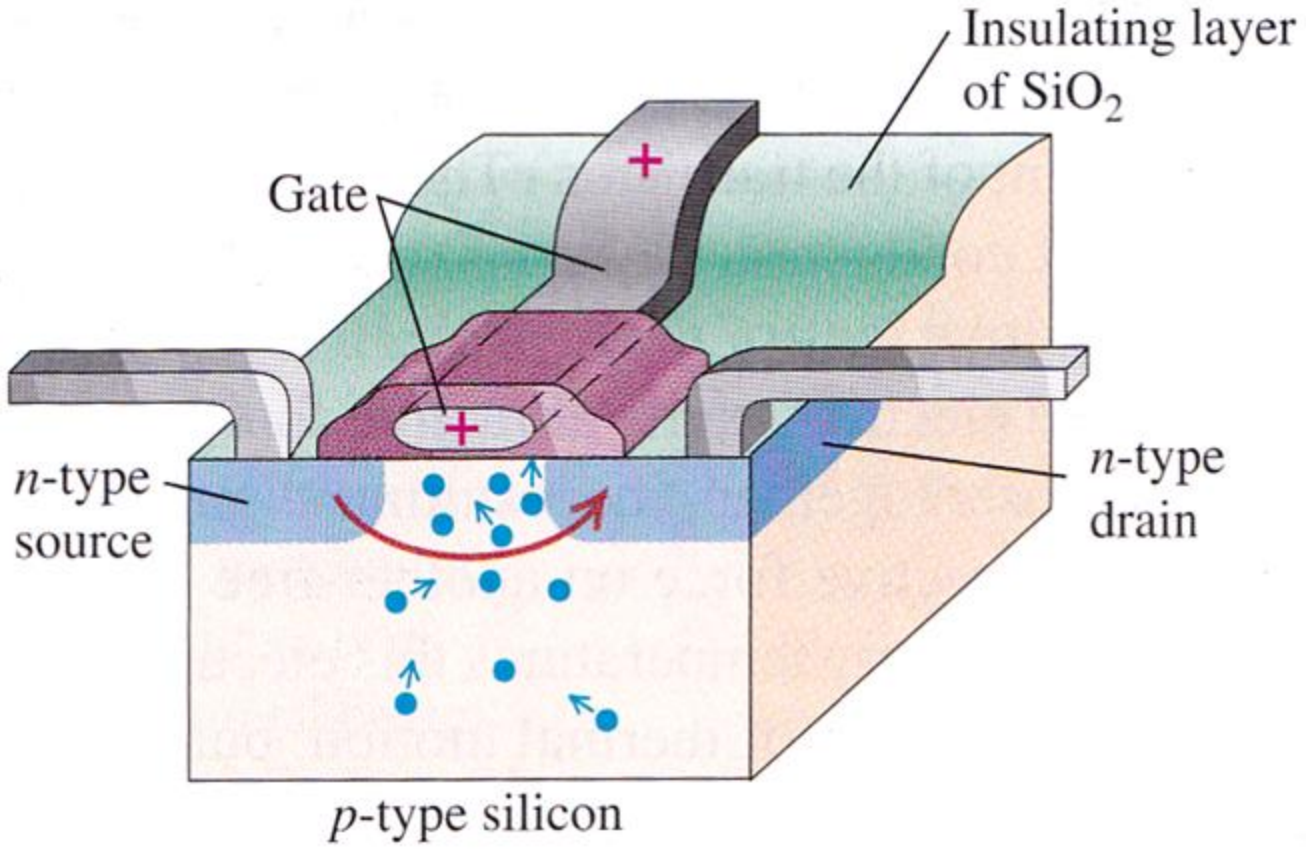


https://madpcb.com

MOSFET-transistor

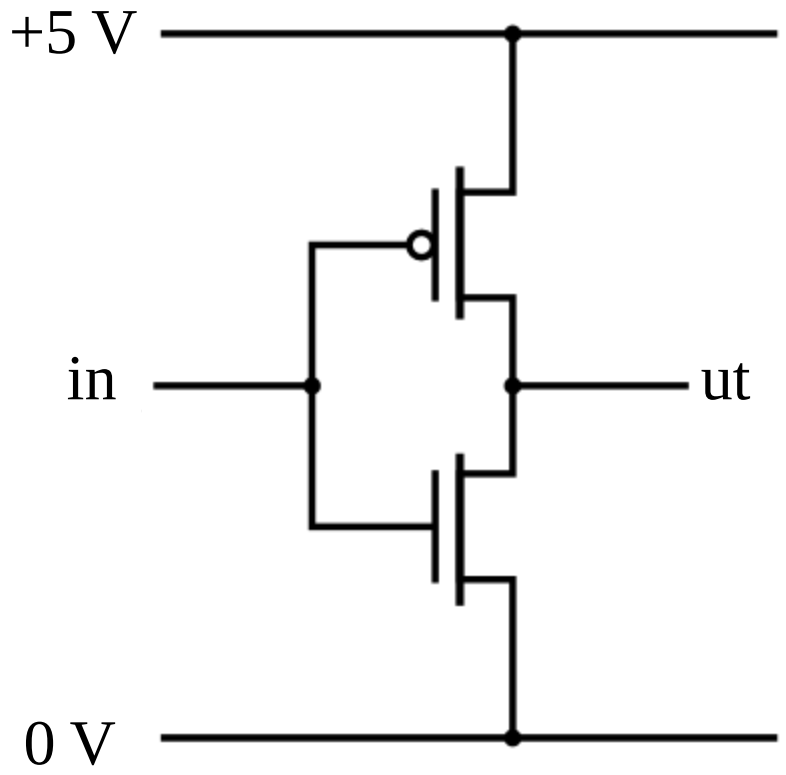
Metal oxide semiconductor field effect transistor

Kallas även MOS



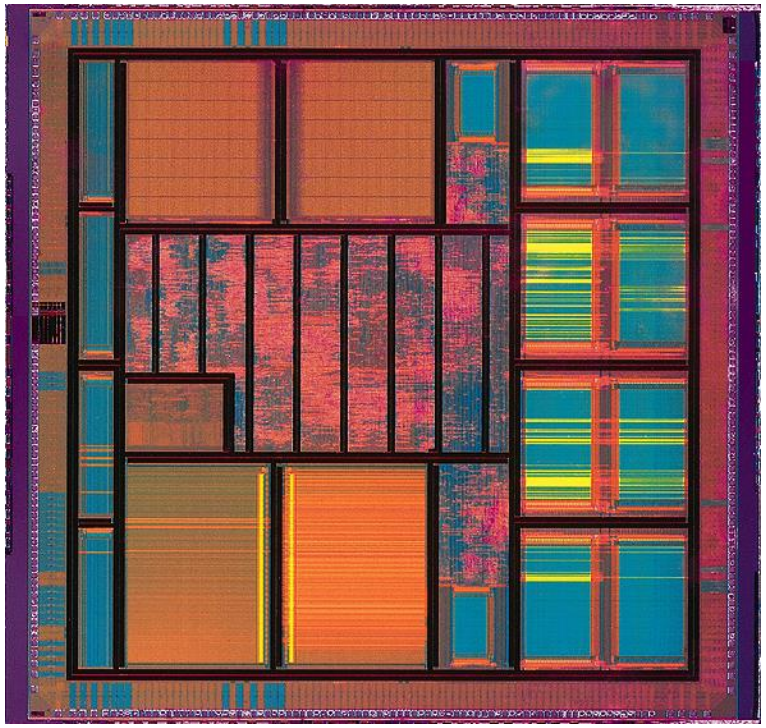
Young and Freedman: University Physics

CMOS (complementary MOS)



Integrerad krets

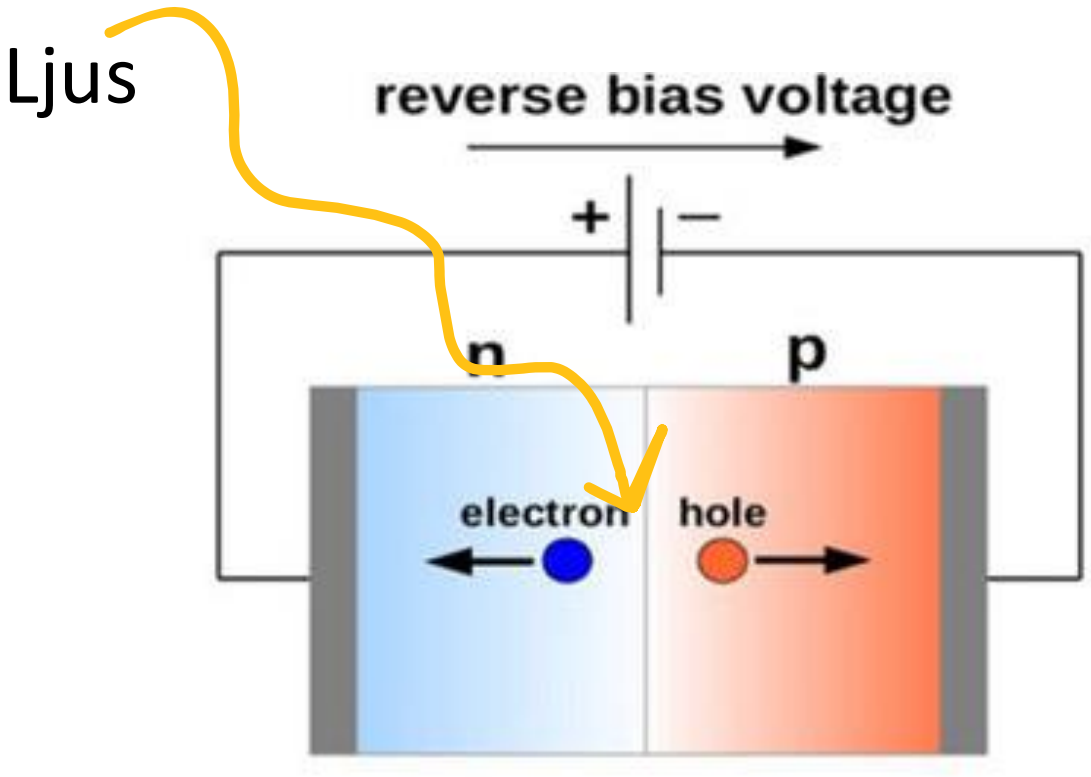
Atmel Diopsis 740

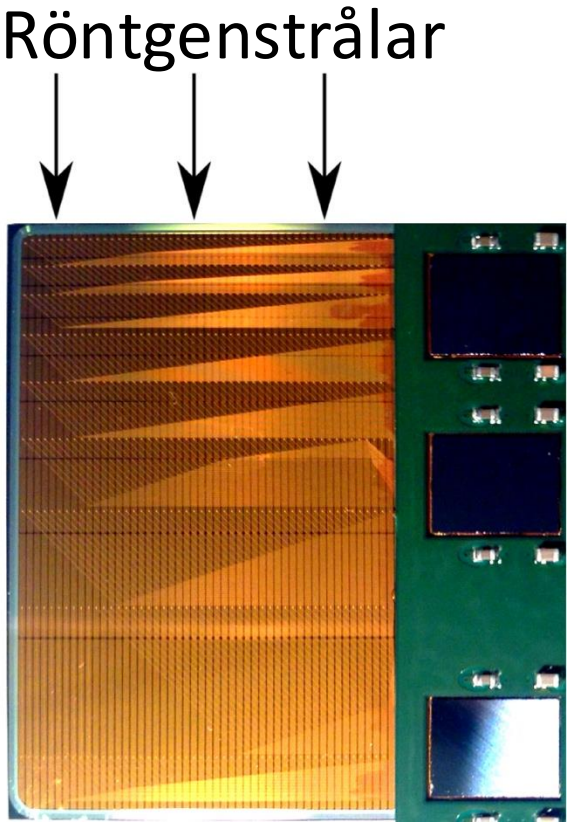


https://en.wikipedia.org/wiki/Die_(integrated_circuit)#/media/File:Diopsis.jpg

Källa: https://en.wikipedia.org/wiki/CMOS

Fotodiod





Lecoq et al.
https://link.springer.com/article/10.1140%2Fepjp%2Fs13360-021-01183-8

Supraledare

Elektroner parar ihop sig två och två
(Cooper par)
Parvis tar spin ut varandra och elektronerna
(en fermion) blir en boson
Detta möjliggörs genom att elektronerna växelverkar
via kristallgittret (fononer)
Alla bosoner kan vara i samma kvanttillstånd (som fotonerna i en laser)
Alla i fas och kollisioner med kristallgittret undviks

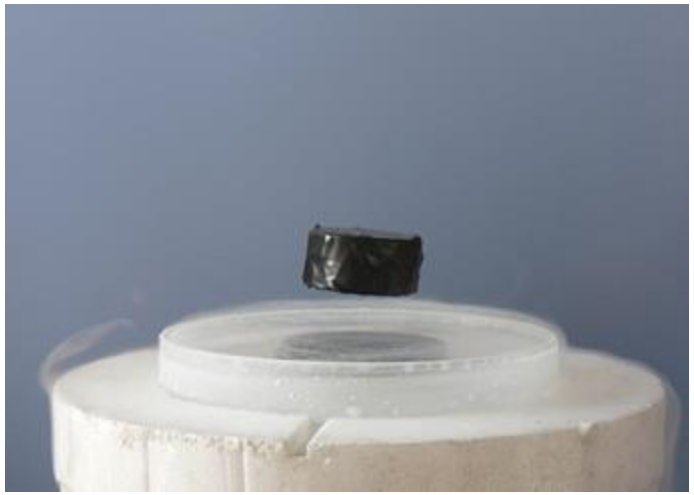
https://en.wikipedia.org/wiki/Superconduc

https://www.youtube.com/watch?v=JIjzJKnpahA

Resistans = 0 för $T < T_C$

För MR används Niobium-Titan med TC = 7 K
Kyls med flytande helium 4.2 K (supraflytande 2.17 K)
För 3T erfordras ca 1000 A

Samma magneter som vid CERN men här kyler man med supraledande helium för att komma upp i 8 T



Supraledare vid rumstemperatur?

Article

Room-temperature superconductivity in a carbonaceous sulfur hydride

https://doi.org/10.1038/s41586-020-2801-z

Received: 21 July 2020

Accepted: 8 September 2020

Published online: 14 October 2020

Elliot Snider[1,6], Nathan Dasenbrock-Gammon[2,6], Raymond McBride[1,6], Mathew Debessai[3], Hiranya Vindana[3], Kevin Vencatasamy[3], Keith V. Lawler[4], Ashkan Salamat[5] & Ranga P. Dias[1,2]✉

One of the long-standing challenges in experimental physics is the observation of room-temperature superconductivity[1,2]. Recently, high-temperature conventional

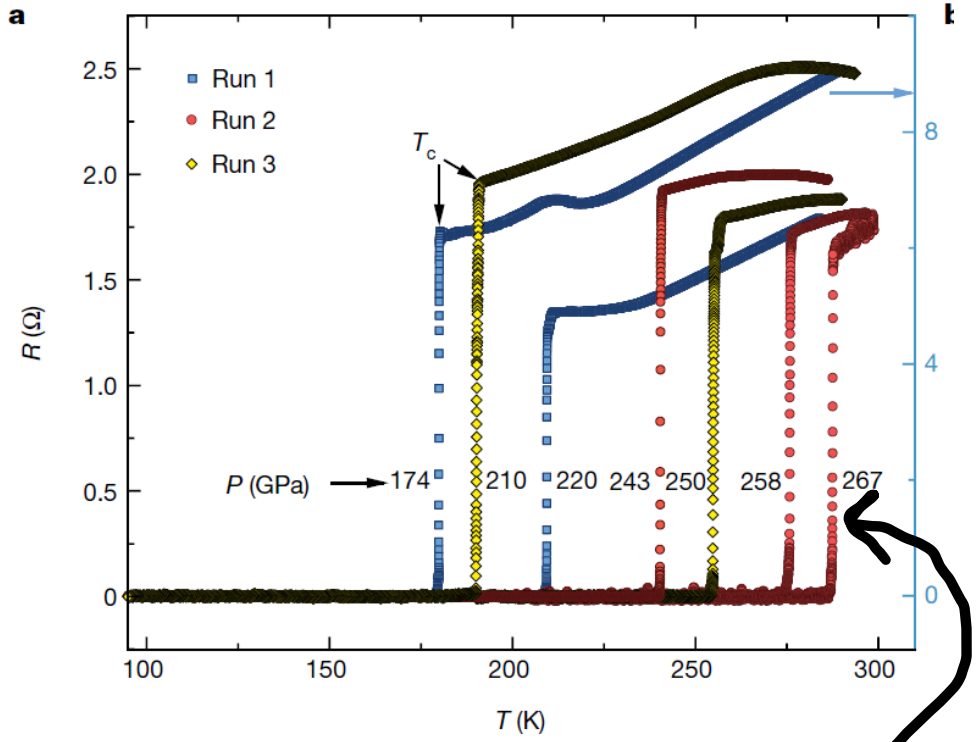
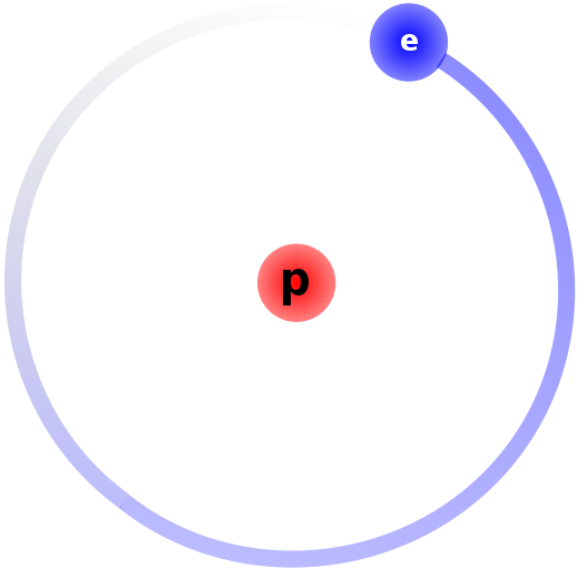
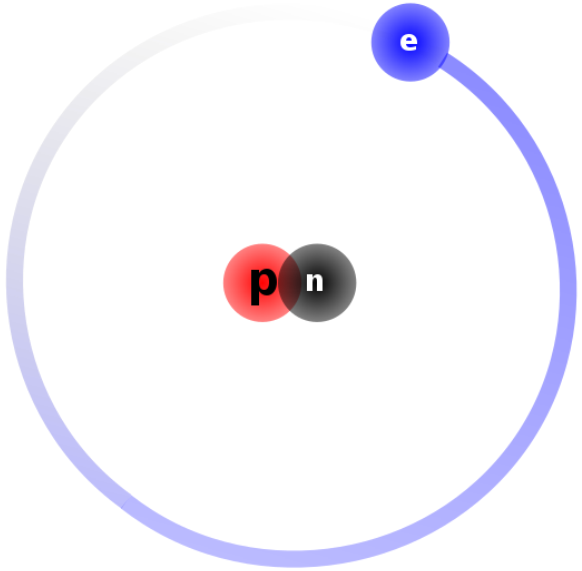
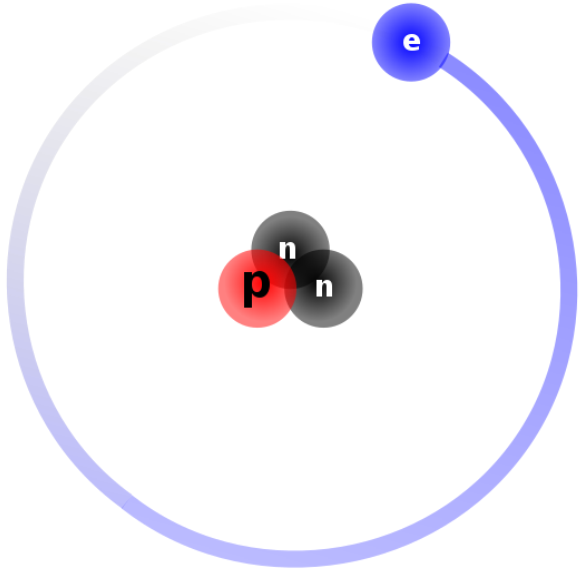


Fig. 1 | Superconductivity in C–S–H at high pressures. a, Temperature-

Kärnfysik







$^{1}_{1}H$

$^{2}_{1}H$

$^{3}_{1}H$

Protium

Deuterium

Tritium

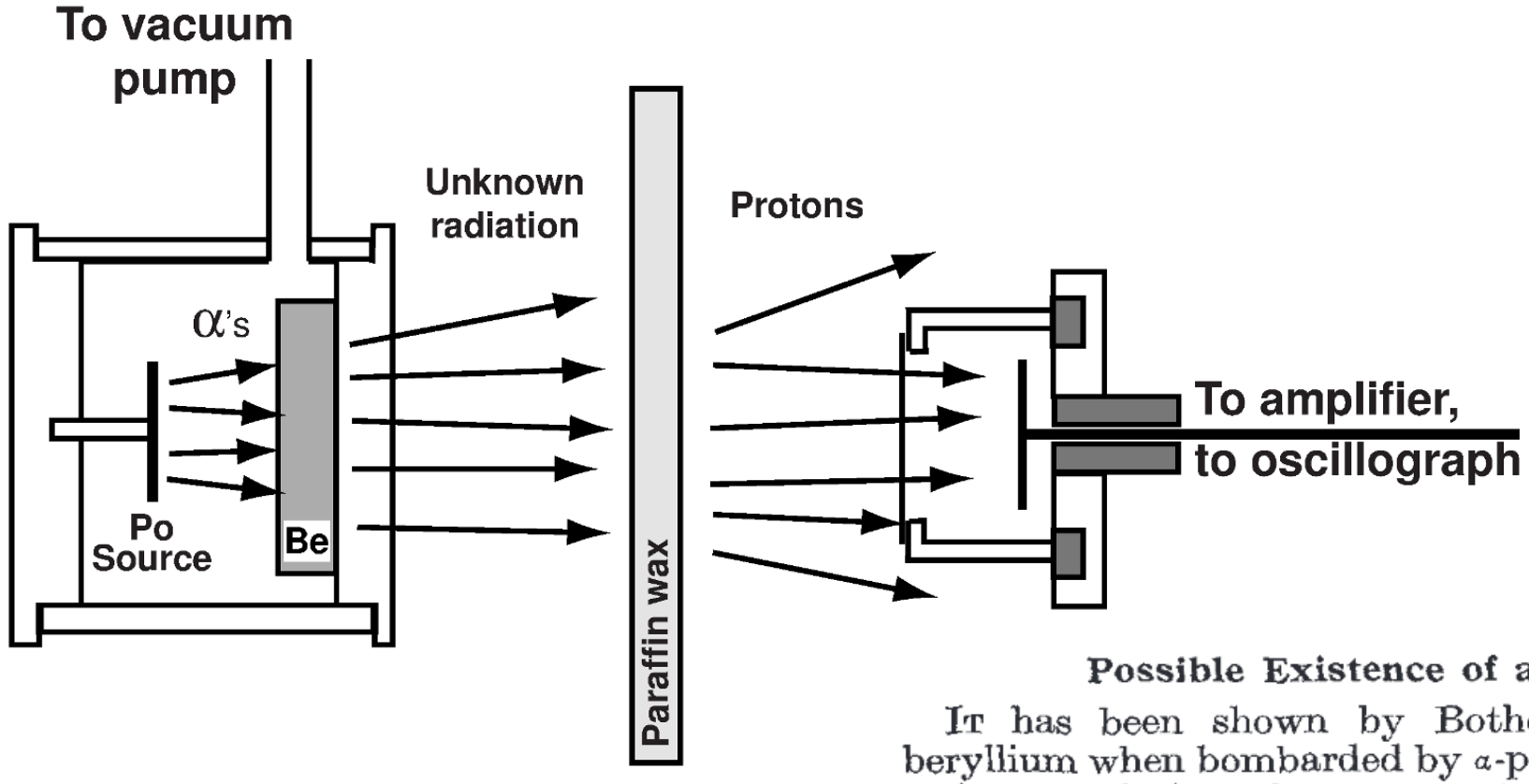
"vanligt väte"

https://en.wikipedia.org/wiki/Isotopes_of_hydrogen#/media/File:Hydrogen_Deuterium_Tritium_Nuclei_Schmatic-en.svg

Isotoper - atomer med samma atomnummer men olika massa

Nuklid	Kärnladdning	Massa (u)
^{1}H	1 e	1,008
^{2}H (deuterium)	1 e	2,014
^{3}H (tritium)	1 e	3,016
^{3}He	2 e	3,016
^{4}He	2 e	4,003
…	…	…
^{16}O	8 e	15,995
^{17}O	8 e	16,999
^{18}O	8 e	17,999

1930-1932 Neutronen upptäcks



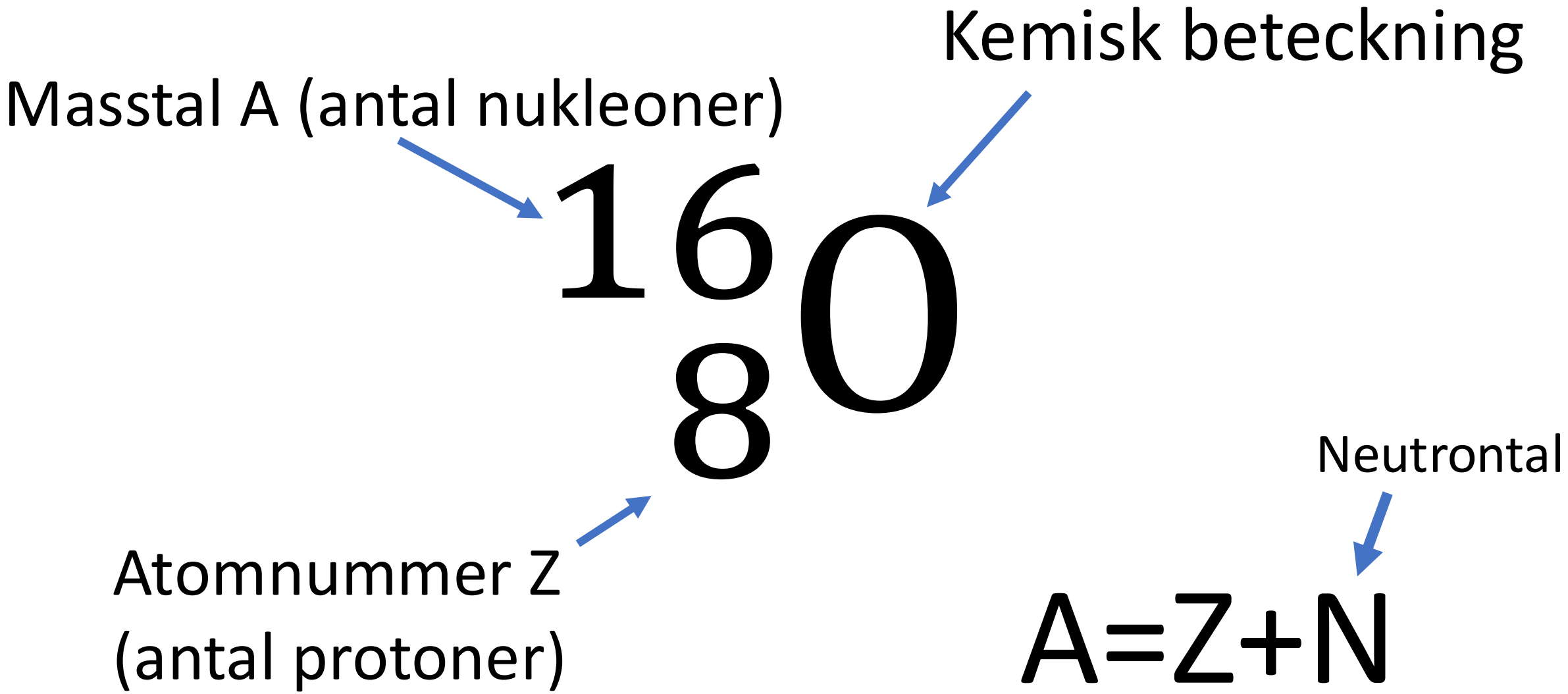
Possible Existence of a Neutron

It has been shown by Bothe and others that beryllium when bombarded by α-particles of polonium emits a radiation of great penetrating power, which has an absorption coefficient in lead of about 0·3 (cm.)$^{-1}$. Recently Mme. Curie-Joliot and M. Joliot found, when measuring the ionisation produced by this beryllium radiation in a vessel with a thin window, that the ionisation increased when matter containing

Bdushaw
https://en.wikipedia.org/wiki/Discovery_of_the_neutron#/media/File:Chadwick_Neutron_Discovery.png

Chadwick, Nature 1932

Beteckning för en nuklid



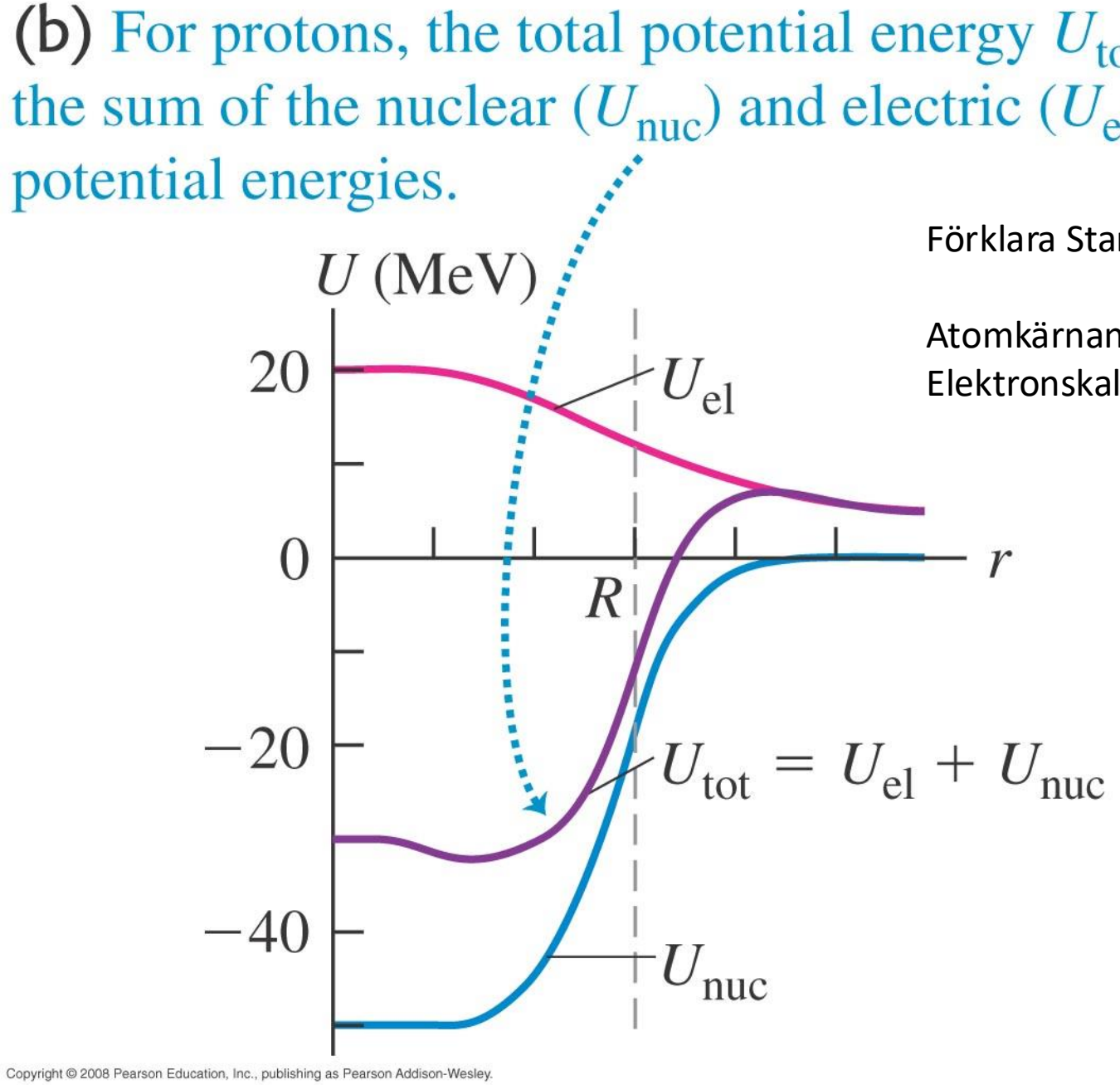
Massenheten 1 u

1 u = 1/12 av massan av en atom av kolisotopen ^{12}C

1 u = $1{,}660539067 \cdot 10^{-27}$ kg
1 u = 931.46 MeV/c2

-protonens massa 938.272 013 MeV/c2
-neutronens massa 939.565560 MeV/c2

Figure 43.3b



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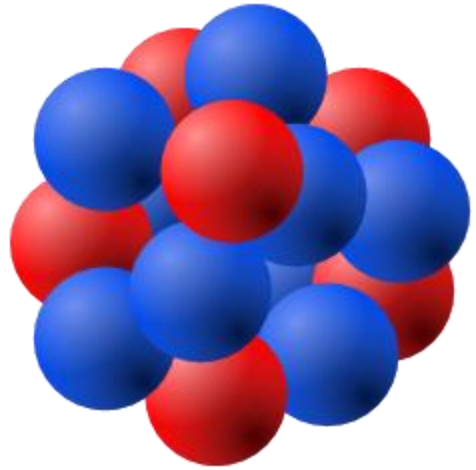
Förklara Stark växelverkan - som binder atomkärnan

Atomkärnan har en skalstruktur som påminner om Elektronskalen i en atom

Kärnradien

$$R = R_0 A^{\frac{1}{3}}$$

Masstal (antal nukleoner)



$$R_0 = 1.2 \cdot 10^{-15}\ m$$

Bindningsenergi

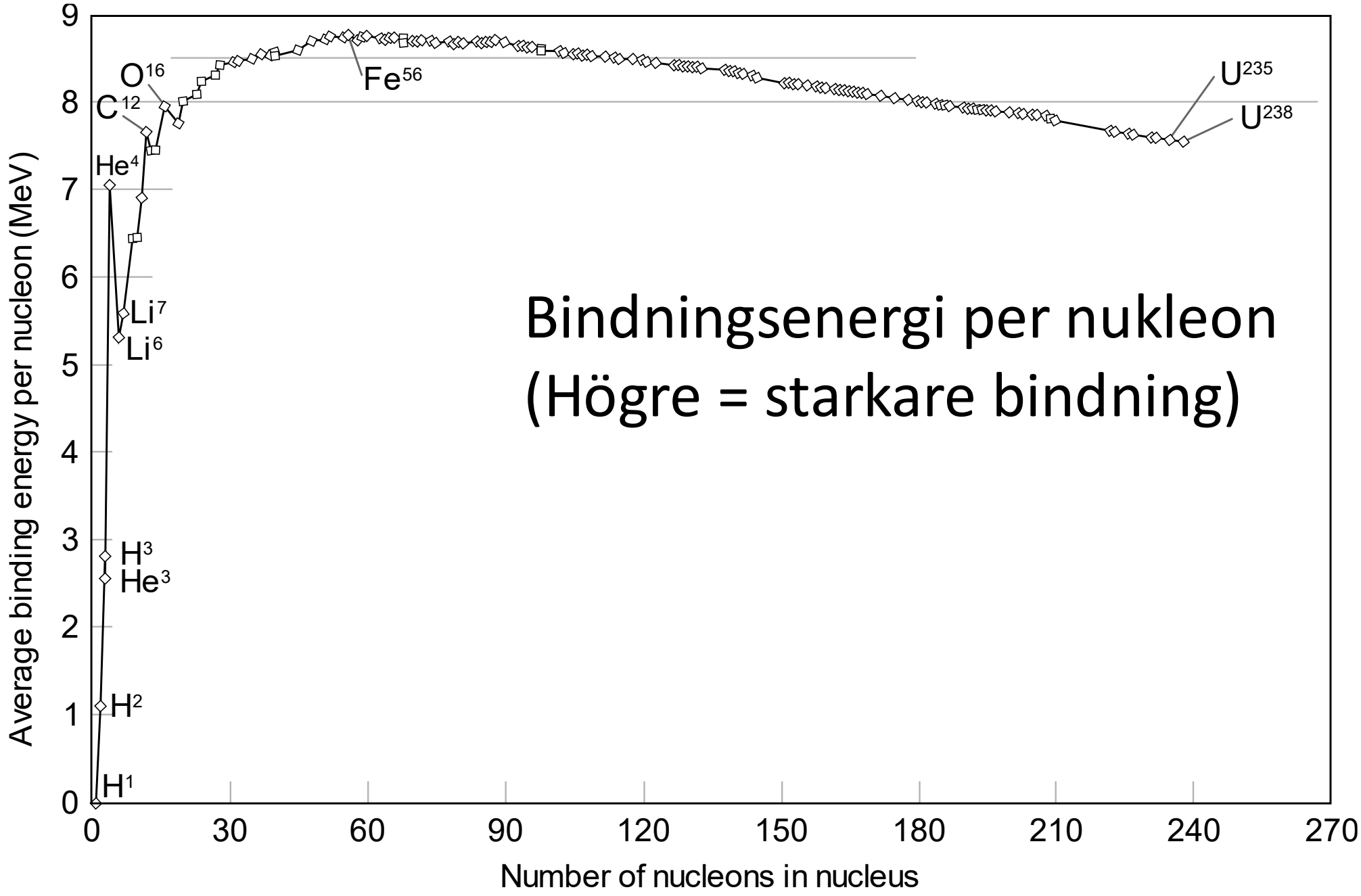
Den energi som måste tillföras för att dela upp en nuklid i Z väteatomer och N neutroner

Väteatomens massa

Neutronens massa

$$E_B = \left(Z M_H + N m_n - {}^{A}_{Z}M\right)c^2$$

Nuklidens massa



Bindningsenergi per nukleon
(Högre = starkare bindning)

FastFission https://en.wikipedia.org/wiki/Nuclear_binding_energy#/media/File:Binding_energy_curve_-_common_isotopes.svg

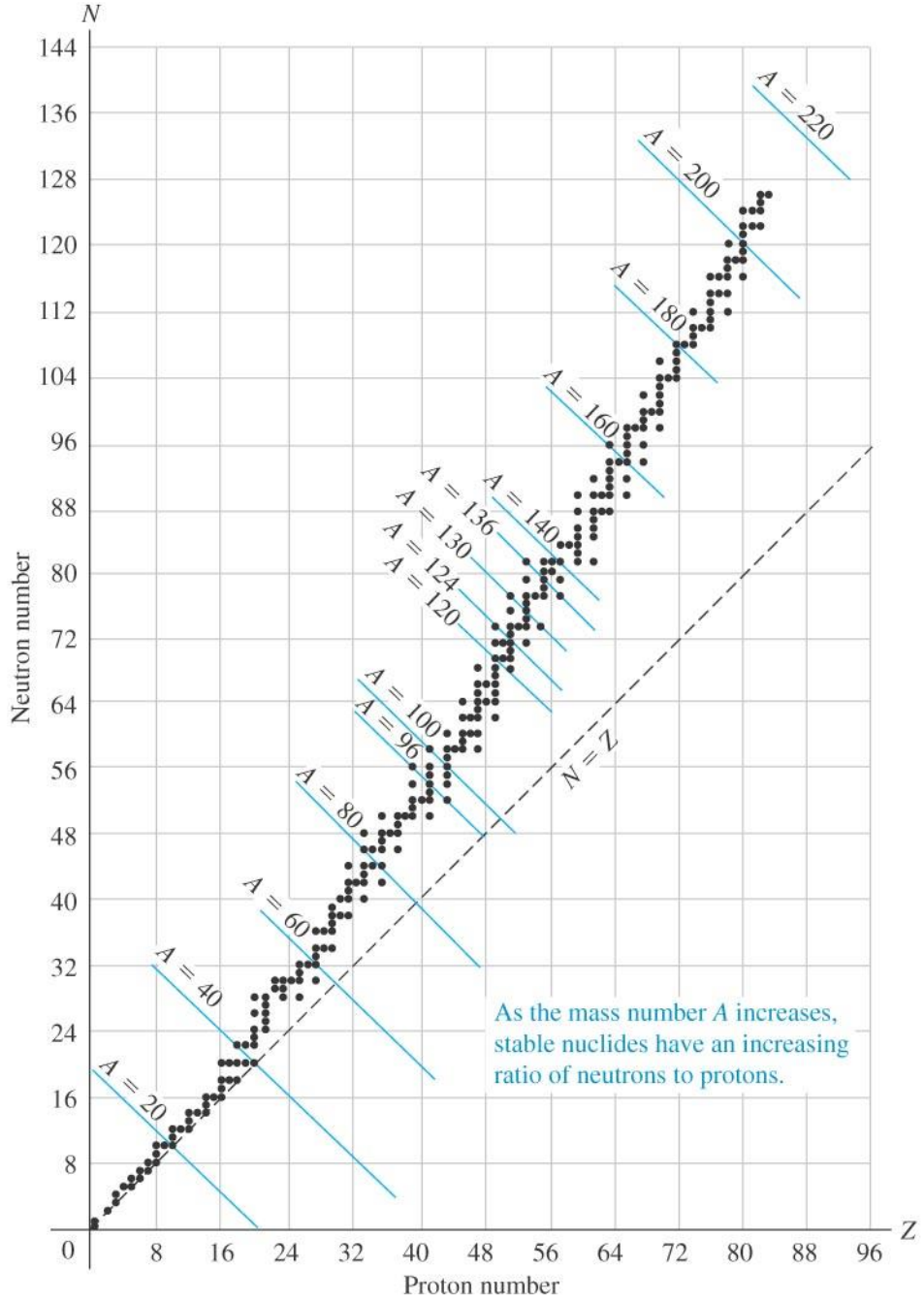
Table 43.2 Neutral Atomic Masses for Some Light Nuclides

Element and Isotope	Atomic Number, Z	Neutron Number, N	Atomic Mass (u)	Mass Number, A
Hydrogen $\left({}_{1}^{1}\mathrm{H}\right)$	1	0	1.007825	1
Deuterium $\left({}_{1}^{2}\mathrm{H}\right)$	1	1	2.014102	2
Tritium $\left({}_{1}^{3}\mathrm{H}\right)$	1	2	3.016049	3
Helium $\left({}_{2}^{3}\mathrm{He}\right)$	2	1	3.016029	3
Helium $\left({}_{2}^{4}\mathrm{He}\right)$	2	2	4.002603	4
Lithium $\left({}_{3}^{6}\mathrm{Li}\right)$	3	3	6.015122	6
Lithium $\left({}_{3}^{7}\mathrm{Li}\right)$	3	4	7.016004	7
Beryllium $\left({}_{4}^{9}\mathrm{Be}\right)$	4	5	9.012182	9
Boron $\left({}_{5}^{10}\mathrm{B}\right)$	5	5	10.012937	10
Boron $\left({}_{5}^{11}\mathrm{B}\right)$	5	6	11.009305	11
Carbon $\left({}_{6}^{12}\mathrm{C}\right)$	6	6	12.000000	12
Carbon $\left({}_{6}^{13}\mathrm{C}\right)$	6	7	13.003355	13
Nitrogen $\left({}_{7}^{14}\mathrm{N}\right)$	7	7	14.003074	14
Nitrogen $\left({}_{7}^{15}\mathrm{N}\right)$	7	8	15.000109	15
Oxygen $\left({}_{8}^{16}\mathrm{O}\right)$	8	8	15.994915	16
Oxygen $\left({}_{8}^{17}\mathrm{O}\right)$	8	9	16.999132	17
Oxygen $\left({}_{8}^{18}\mathrm{O}\right)$	8	10	17.999160	18

Source: A. H. Wapstra and G. Audi, *Nuclear Physics* **A595,** 4 (1995).

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Figure 43.4



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Sönderfall, antal kärnor som är kvar

$$N(t) = N_0 e^{-\lambda t}$$

Halveringstid

$$T_{\text{mean}} = \frac{1}{\lambda} = \frac{T_{1/2}}{\ln 2} = \frac{T_{1/2}}{0.693}$$

(lifetime T_{mean}, decay constant λ, and half-life $T_{1/2}$)